

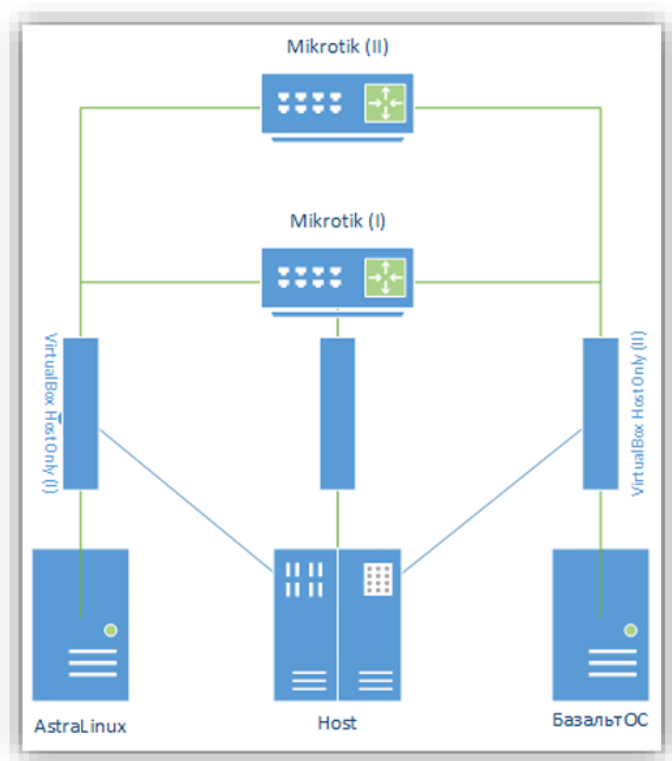
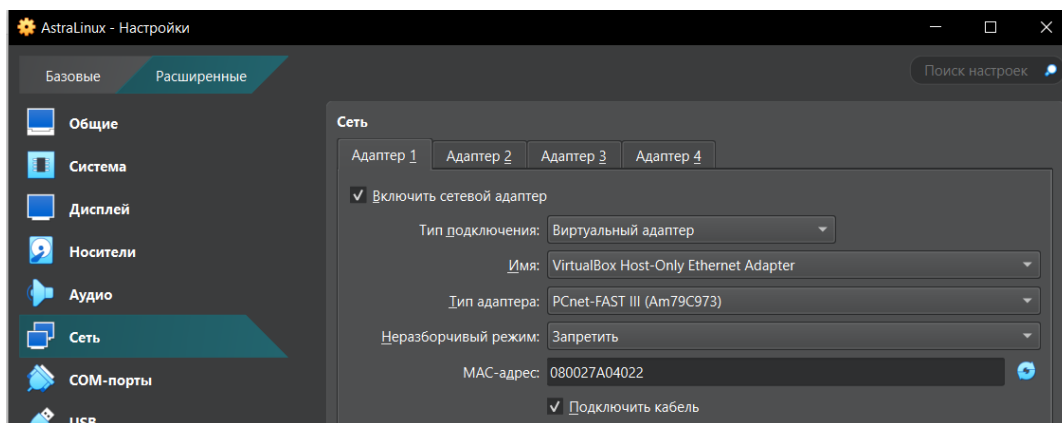


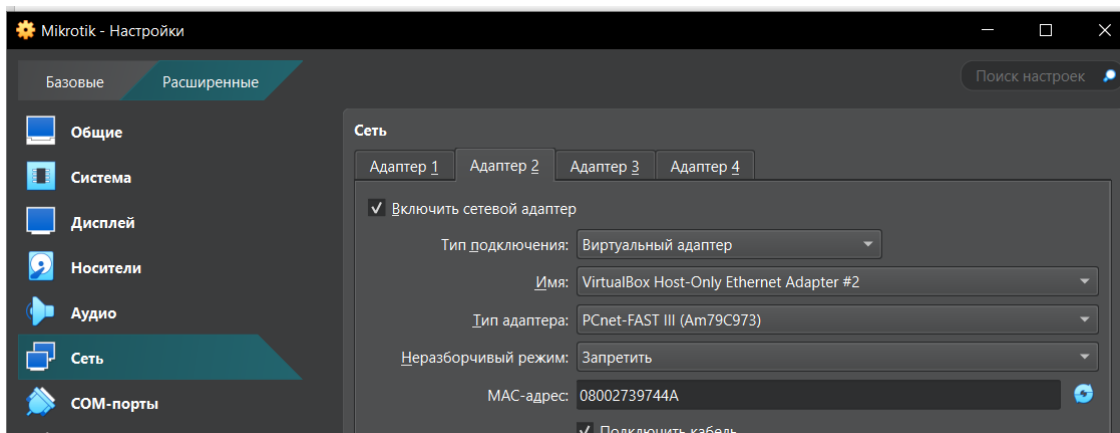
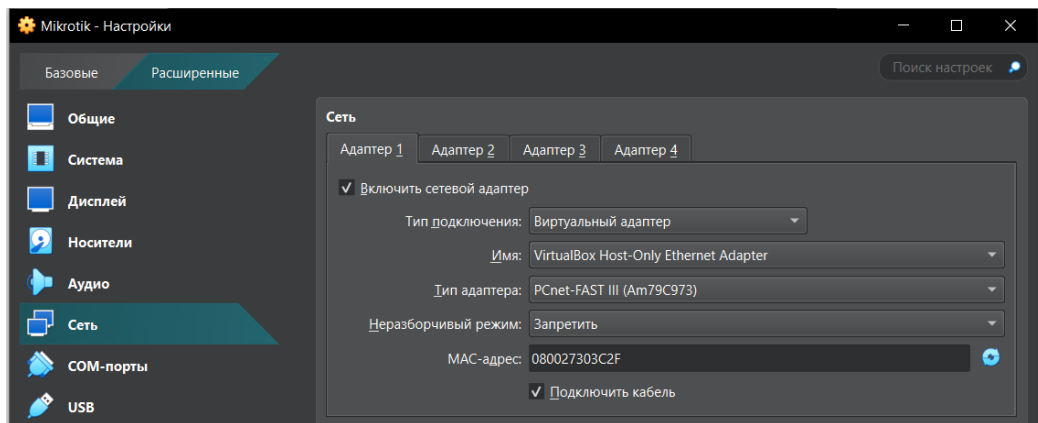
Рисунок 1 – Конфигурация сети для практического занятия

1. Настройка конфигурации:

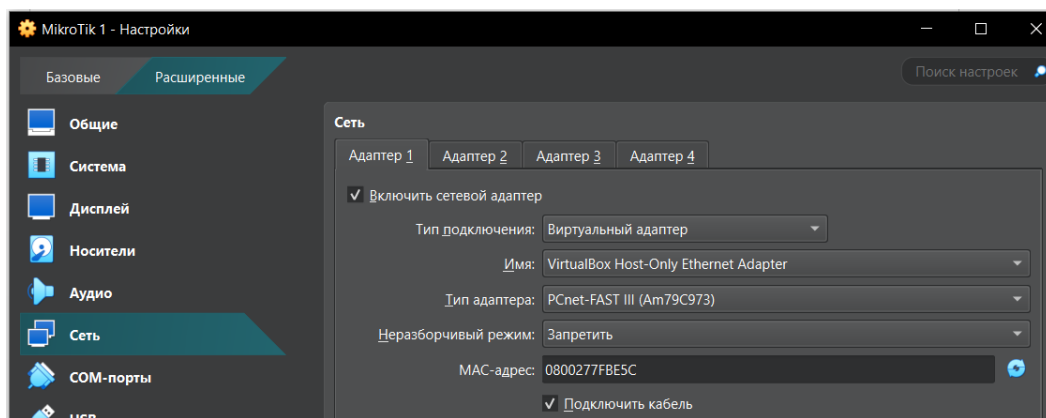
Астра Линукс:

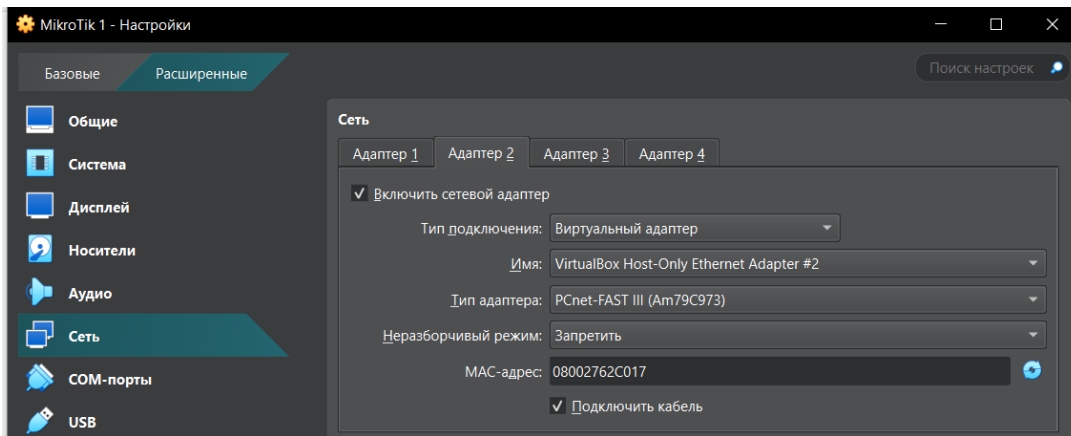


Микротик:

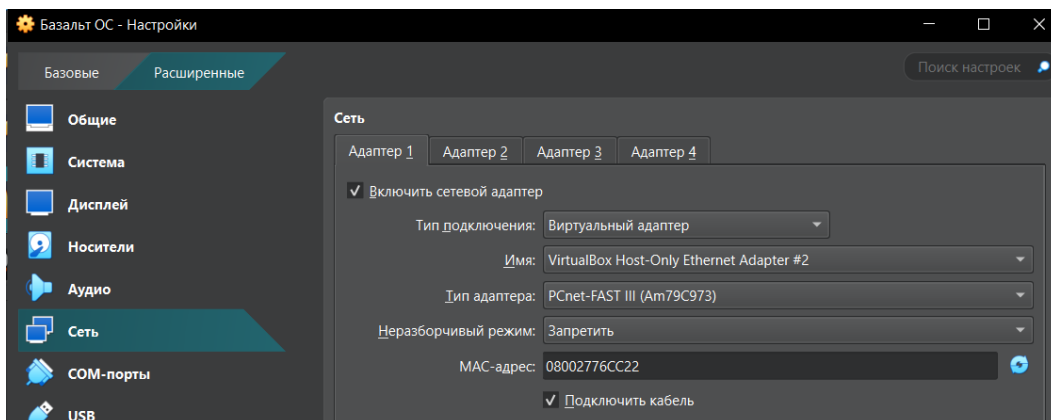


Микротик 1:





БазальтОС:



2.

Исходные данные:

Подсеть 10.10.N.0/24

Разделение подсети:

Первая подсеть: 10.10.16.0/25 (адреса 10.10.16.1–10.10.16.126).

Вторая подсеть: 10.10.16.128/25 (адреса 10.10.16.129–10.10.16.254)

255.255.255.0 (/24)

VirtualBox Host Only 1: 10.10.16.1/25

VirtualBox Host Only 2: 10.10.16.129/25

Виртуальные сети			
Сети NAT			
Облачные сети			
Имя	IPv4 префикс	IPv6 префикс	DHCP сервер
VirtualBox Host-Only Ethernet Adapter	10.10.16.1/24		Выключен
VirtualBox Host-Only Ethernet Adapter #2	10.10.16.129/24		Выключен

Установка адресов на виртуальных машинах:

Астра Линукс:

Адрес: 10.10.16.5/25 на eth0

```
You have new mail.  
user@host-astra:~$ sudo ip addr add 10.10.16.5/25 dev eth0  
user@host-astra:~$ ip a | grep "eth0"  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast  
    link/ether 08:00:27:12:34:56 brd ff:ff:ff:ff:ff:ff  
    inet 169.254.4.217/16 brd 169.254.255.255 scope link eth0:avahi  
    inet 10.10.16.5/25 scope global eth0  
user@host-astra:~$
```

Микротик:

Адрес: 10.10.16.2/25 (ether1)

Адрес: 10.10.16.130/25 (ether2)

```
[admin@MikroTik1] > ip address/add address=10.10.16.2/25  
interface: ether1  
[admin@MikroTik1] > ip address/add address=10.10.16.130/25  
interface: ether2  
[admin@MikroTik1] > ip address/print  
Columns: ADDRESS, NETWORK, INTERFACE  
# ADDRESS NETWORK INTERFACE  
0 10.16.64.4/18 10.16.64.0 ether1  
1 192.168.56.101/24 192.168.56.0 ether2  
2 10.10.16.66/26 10.10.16.64 ether2  
3 10.10.16.10/26 10.10.16.0 ether1  
4 10.10.16.2/25 10.10.16.0 ether1  
5 10.10.16.130/25 10.10.16.128 ether2  
[admin@MikroTik1] >
```

Микротик 1:

Адрес: 10.10.16.3/25 (ether2)

Адрес: 10.10.16.131/25 (ether3)

```
[admin@MikroTik1] > ip address/add address=10.10.16.3/25
interface: ether2
[admin@MikroTik1] > ip address/add address=10.10.16.131/25
interface: ether3
[admin@MikroTik1] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 10.10.16.70/26 10.10.16.64 ether2
1 10.10.16.130/26 10.10.16.128 ether3
2 10.10.16.3/25 10.10.16.0 ether2
3 10.10.16.131/25 10.10.16.128 ether3
[admin@MikroTik1] > _
```

Базальт:

Адрес: 10.10.16.132/24 (enp0s3)

```
[root@host-basalt ~]# sudo ip addr add 10.10.16.132/25 dev enp0s3
[root@host-basalt ~]# ip a | grep "enp0s3"
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    inet 10.16.63.254/18 brd 10.16.63.255 scope global dynamic noprefixroute enp0s3
    inet 10.10.16.132/25 scope global enp0s3
[root@host-basalt ~]#
```

Проверка пинга.

Пинг должен проходить только между адресами, находящимися в одной подсети:

Астра Линукс <=> Микротик:

The image shows two side-by-side terminal windows from Oracle VM VirtualBox. The left window is titled 'AstraLinux (Снимок 1 астра) [Работающий] - Oracle VM VirtualBox' and shows a user at 'user@host-astra' performing a ping test to 10.10.16.2. The output shows 13 successful pings with varying times (e.g., 3.08 ms, 1.46 ms, 1.98 ms, 2.50 ms, 1.79 ms, 1.85 ms, 1.74 ms, 2.19 ms, 2.60 ms, 1.93 ms, 1.06 ms, 1.66 ms). The right window is titled 'Mikrotik [Работающий] - Oracle VM VirtualBox' and shows a user at '[admin@MikroTik1]' performing a ping test to 10.10.16.5. The output shows 9 successful pings with times (e.g., 3ms962us, 2ms462us, 1ms045us, 2ms405us, 2ms263us, 1ms770us, 3ms863us, 1ms942us, 2ms159us). Both windows show network configuration details at the top, including IP addresses and interfaces.

Астра Линукс <=> Микротик1:

Микротик <=> Микротик1:

БазальтОС <=> Микротик:

БазальтОС <=> Микротик1:

3. Создание моста на первом Микротике:

```
Podkorytova Aleksandra IP-311

[admin@MikroTik1] > interface/bridge/add name=bridge1
[admin@MikroTik1] > interface/bridge/print
Flags: X - disabled, R - running
 0 R name="bridge1" mtu=auto actual-mtu=1500 l2mtu=65535 arp=enabled
  arp-timeout=auto mac-address=0E:23:B8:3C:CC:41 protocol-mode=rstp
  fast-forward=yes igmp-snooping=no auto-mac=yes ageing-time=5m
  priority=0x8000 max-message-age=20s forward-delay=15s
  transmit-hold-count=6 vlan-filtering=no dhcp-snooping=no
  port-cost-mode=long mvrp=no max-learned-entries=auto
[admin@MikroTik1] > _
```

Добавление интерфейсов в мост

Состояние порта в bridge

Показывает, какие интерфейсы в мосту, и что с ними:

/interface bridge port print

```
[admin@MikroTik1] > interface/bridge/port add interface=ether1 bridge=bridge1
[admin@MikroTik1] > interface/bridge/port add interface=ether2 bridge=bridge1
[admin@MikroTik1] > interface/bridge/port print
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, HORIZON
# INTERFACE BRIDGE HW PVID PRIORITY HORIZON
0 ether1 bridge1 yes 1 0x80 none
1 ether2 bridge1 yes 1 0x80 none
[admin@MikroTik1] >
```

- **INTERFACE:** Имя интерфейса (ether1, ether2).
- **BRIDGE:** Имя моста (bridge1).
- **HW:** yes — аппаратная поддержка моста (корректно).
- **PVID:** Уникальный идентификатор порта (1 для обоих — нормально).
- **PRIORITY:** Приоритет порта (0x00 — значение по умолчанию).
- **HORIZON:** none — горизонт петли не настроен (по умолчанию).

```
[admin@MikroTik1] > ip address/print
Flags: S - SLAVE
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 S 10.10.16.130/25 10.10.16.128 ether2
1 S 10.10.16.2/25 10.10.16.0 ether1
[admin@MikroTik1] > ip address/remove 0,1
[admin@MikroTik1] > ip address/add address=10.10.16.130/25
interface: bridge1
[admin@MikroTik1] > ip address/add address=10.10.16.2/25
interface:
bridge1 ether1 ether2 lo
interface: bridge1
[admin@MikroTik1] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 10.10.16.130/25 10.10.16.128 bridge1
1 10.10.16.2/25 10.10.16.0 bridge1
[admin@MikroTik1] >
```

Статистика по самому bridge-интерфейсу:

```

[admin@MikroTik1] > interface/bridge/monitor br
bridge1      numbers
[admin@MikroTik1] > interface/bridge/monitor bridge1
state: enabled
current-mac-address: 08:00:27:30:3C:2F
root-bridge: yes
root-bridge-id: 0x8000.08:00:27:30:3C:2F
root-path-cost: 0
root-port: none
port-count: 2
designated-port-count: 2
fast-forward: yes
-- [Q quit|D dump|C-z pause]

```

Показывает **текущее состояние моста**, в том числе — **информацию STP** (если включён).

State: enabled — мост активен

root-bridge-id — MAC и приоритет корневого моста (тот, кто выиграл STP)

root-path-cost — суммарная стоимость пути до root bridge

Таблица MAC-адресов:

```

[admin@MikroTik1] > interface/bridge/host/print
Flags: D - DYNAMIC; L - LOCAL
Columns: MAC-ADDRESS, ON-INTERFACE, BRIDGE
#   MAC-ADDRESS      ON-INTERFACE  BRIDGE
0 DL 08:00:27:30:3C:2F bridge1      bridge1
1 DL 08:00:27:39:74:4A ether2        bridge1
2 D  08:00:27:62:C0:17 ether2        bridge1
3 D  08:00:27:76:CC:22 ether2        bridge1
4 D  08:00:27:7F:BE:5C ether1        bridge1
5 D  08:00:27:A0:40:22 ether1        bridge1
6 D  0A:00:27:00:00:05 ether1        bridge1
7 D  0A:00:27:00:00:0D ether2        bridge1
[admin@MikroTik1] >

```

- **MAC-ADDRESS** — какие устройства находятся в сети
- **ON-INTERFACE** — через какой интерфейс они были замечены

Показывает в **реальном времени**, какой трафик проходит через мост (или любой другой интерфейс).

rx / tx packets — сколько пакетов принял/отправил

rx / tx bits-per-second — объём трафика в реальном времени

The screenshot shows a terminal window with two panes. The left pane displays the output of a ping command from 10.10.16.3 to 10.10.16.2, showing successful transmission with 0% packet loss. The right pane shows the output of the 'show interface' command for 'bridge1', displaying statistics for rx-packets-per-second, tx-packets-per-second, and other metrics. Below this, a packet capture analysis for 'Frame 1' is shown, detailing its structure as an IEEE 802.3 Ethernet frame with a Logical-Link Control (LLC) sublayer.

```
AstraLinux (Снимок 1 астра) [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка
[1]+  Остановлен  sudo nano /etc/sysctl.conf
user@host-astra:~$ ping 10.10.16.3
PING 10.10.16.3 (10.10.16.3) 56(84) bytes of data:
^C
--- 10.10.16.3 ping statistics ---
2189 packets transmitted, 0 received, 100% packet loss, time 2240495ms

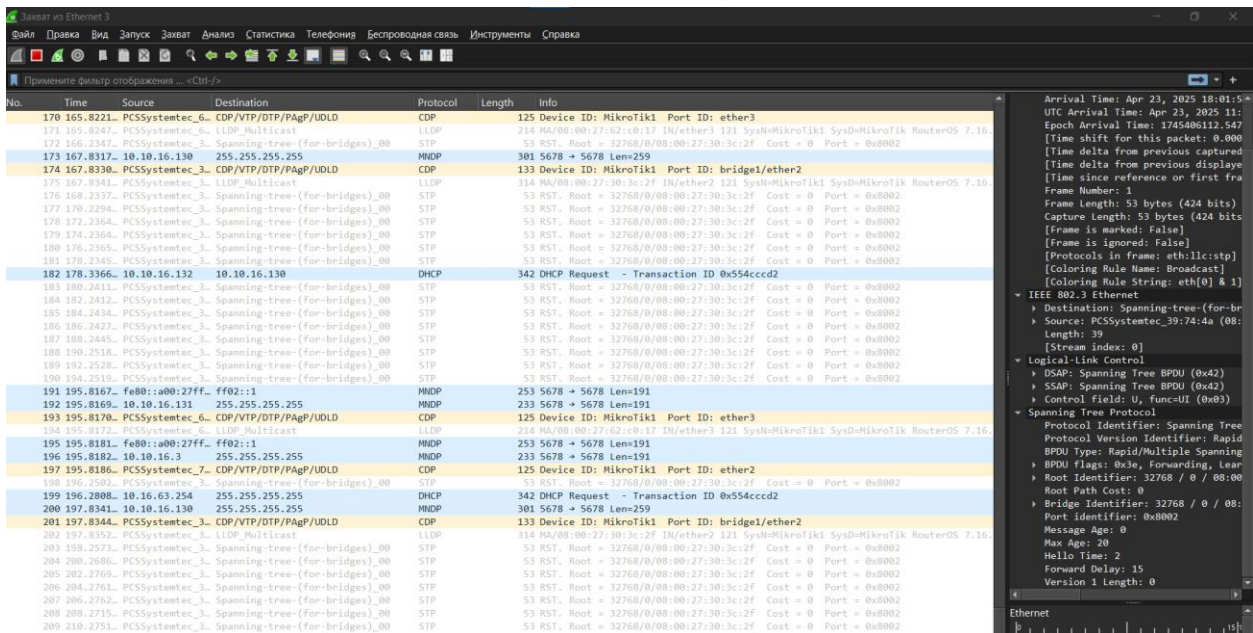
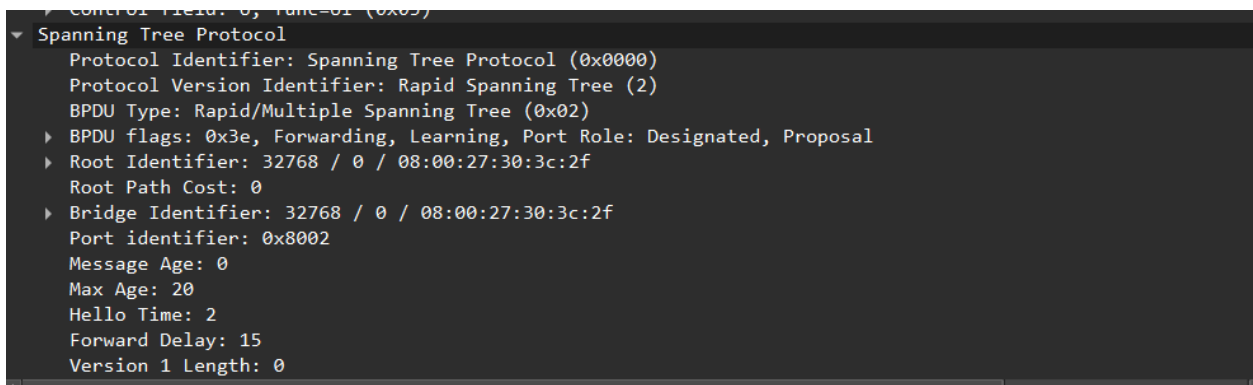
user@host-astra:~$ ping 10.10.16.2
PING 10.10.16.2 (10.10.16.2) 56(84) bytes of data:
64 bytes from 10.10.16.2: icmp_seq=1 ttl=64 time=2.36 ms
64 bytes from 10.10.16.2: icmp_seq=2 ttl=64 time=0.843 ms
64 bytes from 10.10.16.2: icmp_seq=3 ttl=64 time=2.00 ms
64 bytes from 10.10.16.2: icmp_seq=4 ttl=64 time=0.703 ms
64 bytes from 10.10.16.2: icmp_seq=5 ttl=64 time=1.40 ms
64 bytes from 10.10.16.2: icmp_seq=6 ttl=64 time=1.13 ms
64 bytes from 10.10.16.2: icmp_seq=7 ttl=64 time=1.00 ms
^C
--- 10.10.16.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6013ms
rtt min/avg/max/mdev = 0.703/1.351/2.363/0.574 ms

0 DL 08:00:27:30:3C:2F bridge1 bridge1
1 DL 08:00:27:39:74:4a ether2 bridge1
2 D 08:00:27:62:C0:17 ether2 bridge1
3 D 08:00:27:76:C0:22 ether2 bridge1
4 D 08:00:27:7F:BE:5C ether1 bridge1
5 D 08:00:27:A0:40:22 ether1 bridge1
6 D 0A:00:27:00:00:05 ether1 bridge1
7 D 0A:00:27:00:00:0D ether2 bridge1
[admin@Mikrotik1] > interface/monitor-traffic interface=bridge1
name: bridge1
rx-packets-per-second: 0
rx-bits-per-second: 0bps
fp-rx-packets-per-second: 0
fp-rx-bits-per-second: 0bps
rx-drops-per-second: 0
rx-errors-per-second: 0
tx-packets-per-second: 0
tx-bits-per-second: 0bps
fp-tx-packets-per-second: 0
fp-tx-bits-per-second: 0bps
tx-drops-per-second: 0
tx-queue-drops-per-second: 0
tx-errors-per-second: 0
-- [Q quitID dumpIC-z pause]
```

4.

The screenshot shows a detailed packet capture analysis for 'Frame 1'. It lists the interface ID, encapsulation type, arrival time, and frame length. The frame is identified as an IEEE 802.3 Ethernet frame with a Logical-Link Control (LLC) sublayer. The destination is 'Spanning-tree-(for-bridges)_00' and the source is 'PCSSystemtec_39'. The frame is marked as 'eth:llc:stp' and 'Broadcast'.

```
▼ Frame 1: 53 bytes on wire (424 bits), 53 bytes captured (424 bits) on interface \Device\NPF_{4432D551-5CD8-4038-AD26-15E0E614E9A5}
  Section number: 1
    ▶ Interface id: 0 (\Device\NPF_{4432D551-5CD8-4038-AD26-15E0E614E9A5})
      Encapsulation type: Ethernet (1)
      Arrival Time: Apr 23, 2025 18:01:52.547780000 Новосибирское стандартное время
      UTC Arrival Time: Apr 23, 2025 11:01:52.547780000 UTC
      Epoch Arrival Time: 1745406112.547780000
      [Time shift for this packet: 0.000000000 seconds]
      [Time delta from previous captured frame: 0.000000000 seconds]
      [Time delta from previous displayed frame: 0.000000000 seconds]
      [Time since reference or first frame: 0.000000000 seconds]
      Frame Number: 1
      Frame Length: 53 bytes (424 bits)
      Capture Length: 53 bytes (424 bits)
      [Frame is marked: False]
      [Frame is ignored: False]
      [Protocols in frame: eth:llc:stp]
      [Coloring Rule Name: Broadcast]
      [Coloring Rule String: eth[0] & 1]
    ▼ IEEE 802.3 Ethernet
      ▶ Destination: Spanning-tree-(for-bridges)_00 (01:80:c2:00:00:00)
      ▶ Source: PCSSystemtec_39:74:4a (08:00:27:39:74:4a)
      Length: 39
      [Stream index: 0]
    ▼ Logical-Link Control
      ▶ DSAP: Spanning Tree BPDU (0x42)
      ▶ SSAP: Spanning Tree BPDU (0x42)
      ▶ Control field: U, func=UI (0x03)
```



Из Wireshark видны пакеты протокола STP (Spanning Tree Protocol) — они идут от MikroTik (с MAC-адресом 08:00:27:e5:b1:10), это говорит о следующем:

Тип трафика: STP (RSTP — Rapid Spanning Tree)

Отправитель: MAC 08:00:27:e5:b1:10 — это Микротик1

Root = 32768/08:00:27:e5:b1:10 — этот MikroTik сам себя считает **корневым мостом**

Port = 0x8002 и т.д. — это **ID порта**, участвующего в STP

Cost = 0 — путь до корня (самого себя) = 0

Что это значит:

- 1) **STP работает корректно** – MikroTik1 рассылает **BPDU-пакеты** (Bridge Protocol Data Units)
- 2) **MikroTik-02 объявляет себя Root Bridge** – потому что он не получает BPDU от других мостов с меньшим приоритетом
- 3) Wireshark **фиксирует STP-трафик на хосте**, значит мост передаёт его на интерфейс vboxnet2

```
Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Rapid Spanning Tree (2)
  BPDU Type: Rapid/Multiple Spanning Tree (0x02)
  ▶ BPDU flags: 0x3e, Forwarding, Learning, Port Role: Designated, Proposal
  ▼ Root Identifier: 32768 / 0 / 08:00:27:30:3c:2f
    Root Bridge Priority: 32768
    Root Bridge System ID Extension: 0
    Root Bridge System ID: PCSSystemtec_30:3c:2f (08:00:27:30:3c:2f)
    Root Path Cost: 0
  ▼ Bridge Identifier: 32768 / 0 / 08:00:27:30:3c:2f
    Bridge Priority: 32768
    Bridge System ID Extension: 0
    Bridge System ID: PCSSystemtec_30:3c:2f (08:00:27:30:3c:2f)
  Port identifier: 0x8002
```

- **Root Identifier** – это Root ID
- **Bridge Identifier** – это Sender ID

5. На Микротике1 (это мой второй микротик):

```
[admin@MikroTik1] > interface/bridge/add name=bridge2 protocol-mode=stp
[admin@MikroTik1] > interface/bridge/port add interface=ether1 bridge=bridge2
invalid value for argument interface:
  input does not match any value of interface
  input does not match any value of interface-list
[admin@MikroTik1] > interface/bridge/port add interface=ether2 bridge=bridge2
[admin@MikroTik1] > interface/bridge/port add interface=ether3 bridge=bridge2
[admin@MikroTik1] > interface/bridge/print
Flags: X - disabled, R - running
 0 R name="bridge2" mtu=auto actual-mtu=1500 l2mtu=65535 arp=enabled
  arp-timeout=auto mac-address=08:00:27:7F:BE:5C protocol-mode=stp
  fast-forward=yes igmp-snooping=no auto-mac=yes ageing-time=5m
  priority=0x8000 max-message-age=20s forward-delay=15s
  transmit-hold-count=6 vlan-filtering=no dhcp-snooping=no
  port-cost-mode=long mvrp=no max-learned-entries=auto
[admin@MikroTik1] >
```

644	647.8585...	PCSSystemtec_3...	LLDP_Multicast	LLDP	314	MA/08:00:27:30:3c:2f	IN/ether2	121	SysN=MikroTik1	SysD=MikroTik	RouterOS 7.16.2 (stable)	2024-11-26
645	648.8582...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
646	650.8674...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
647	652.8786...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
648	654.8800...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
649	656.8785...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
650	658.8871...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
651	660.8937...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
652	662.8914...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
653	664.8905...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
654	666.8899...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
655	668.8964...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
656	670.8968...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
657	672.8939...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				
658	674.8910...	PCSSystemtec_3...	Spanning-tree-(for-bridges)_00	STP	52	Conf. Root = 32768/0/08:00:27:30:3c:2f	Cost = 0	Port = 0x8002				

▼ Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

▶ BPDU flags: 0x00

▶ Root Identifier: 32768 / 0 / 08:00:27:30:3c:2f

Root Path Cost: 0

▶ Bridge Identifier: 32768 / 0 / 08:00:27:30:3c:2f

Port identifier: 0x8001

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

Протокол STP работает на обоих Микротиках.